To design in detail the many complex pieces of such interferometers, to make them all fit together and work together properly, and to keep costs under control and bring the interferometers to completion within a reasonable time require a different culture: a culture of tight coordination, with subgroups of each team focusing on well-defined tasks and a single director making decisions about what tasks will be done when and by whom.

The road from freewheeling independence to tight coordination is a painful one. The world's biology community is traveling that road, with cries of anguish along the way, as it moves toward sequencing the human genome. And we gravitational-wave physicists have been traveling that road since 1984, with no less pain and anguish. I am confident, however, that the excitement, pleasure, and scientific payoff of detecting the waves and deciphering their symphonies will one day make the pain and anguish fade in our memories.

The first sharp turn on our painful road was a 1984 shotgun marriage between the Caltech and MIT teams—each of which by then had about eight members. Richard Isaacson of the D.S. National Science Foundation (NSF) held the shotgun and demanded, as the price of the taxpayers' financial support, a tight marriage in which Caltech and MIT scientists jointly developed the interferometers. Drever (resisting like mad) and Weiss (willingly accepting the inevitable) said their vows, and I became the marriage counselor, the man with the task of forging consensus when Drever pulled in one direction and Weiss in another. It was a rocky marriage, emotionally draining for all; but gradually we began to work together.

The second sharp turn came in November 1986. A committee of eminent physicists—experts in all the technologies we need and experts in the organization and management of large scientific projects—spent an entire week with us, scrutinizing our progress and plans, and then reported to NSF. Our progress got high marks, our plans got high marks, and our prospects for success—for detecting waves and deciphering their symphonies—were rated as high. But our culture was rated as awful; we were still immersed in the loosely knit, freewheeling culture of our birth, and we could never succeed that way, NSF was told. Replace the Drever—Weiss—Thorne troika by a single director, the committee insisted—a director who can mold talented individualists into a tightly knit and